

Painting Your Image

The next step is to prepare an image by setting a portion of it to the RGB (red, green, blue) color value (255, 0, 255). Computer graphics often use magenta to mark which pixels of an image should be rendered as transparent. Our program will treat these magenta pixels like a green screen in video production, replacing them with a resized version of the initial image. Of course, this resized image will have its own smaller magenta area, which the program will replace with another resized image. The base case occurs when the final image has no more magenta pixels, at which point the algorithm is done.

[*Figure 14-2*](#) shows the progression of images created as the resized image is recursively applied to the magenta pixels. In this example, a model stands in front of an art museum exhibit that has been replaced with magenta pixels, turning the photograph itself into the exhibit. You can download this base image from <https://inventwithpython.com/museum.png>.

Be sure you use only the pure (255, 0, 255) magenta color for painting the magenta area in your image. Some tools may have a fading effect that produces a more natural look. For example, Photoshop's Brush tool will produce faded magenta pixels on the outline of the painted area, so you will need to use the Pencil tool instead, which paints using only the precise pure magenta color you've selected. If your graphics program doesn't allow you to specify the precise RGB color for drawing, you can copy and paste the colors from the PNG image at <https://inventwithpython.com/magenta.png>.

The magenta area in the image can be of any size or shape; it does not have to be an exact, contiguous rectangle. You can see that in [*Figure 14-2*](#), the museum visitor cuts into the magenta rectangle, placing them in front of the recursive image.

If you make your own images with Droste Maker, you should use the PNG image file format instead of JPEG. JPEG images use *lossy* compression techniques to keep the file size small that introduce slight imperfections. These are usually imperceptible to the human eye and don't affect overall image quality. However, this lossy compression will replace the pure (255, 0,